



National Institute of
Diabetes and Digestive
and Kidney Diseases

Diabetes & Prediabetes Tests

This content compares the following tests

- [fasting plasma glucose \(FPG\) test](#)
- [oral glucose tolerance test \(OGTT\)](#)
- [A1C test](#)
- [random plasma glucose \(RPG\) test](#)

These tests may be used to screen for and diagnose **diabetes** and to detect individuals with [prediabetes](#).¹

Confirming diagnosis of type 2 diabetes and prediabetes

Diagnosis of type 2 diabetes is definitively made by blood tests. Diagnosis requires two abnormal test results from the same sample or from two different samples. Repeat the test using one of the following methods

- Repeat the same test or run a different test.¹ The second test should be done immediately.¹
- If two test results indicate diabetes, consider the diagnosis confirmed.¹
- If the two different tests conflict, repeat the test that is above the diagnostic threshold. Diagnosis is made based on the confirmed test.¹

If the patient's diabetes test results are close to—but not within—the diagnostic range of the test, the patient may have prediabetes. Health care professionals may wish to observe patients with prediabetes whose condition is likely to progress, recommend lifestyle modifications, and retest in 3 to 6 months.¹



Health care professionals reviewing test results.

Interpreting laboratory results

When interpreting laboratory results, health care professionals should

- consider that all laboratory test results represent a range, rather than an exact number²
- know the A1C assay methods used by their laboratory¹
- send blood samples to be tested by a laboratory that uses an A1C analysis method certified by the NGSP to ensure the results are standardized to the [Diabetes Control and Complications Trial \(DCCT\)](#) assay¹
- consider the possibility of interference in the A1C test when a result is below 4%, above 15%, or is at odds with other diabetes test results¹
- consider each patient's profile, including risk factors and history, and individualize diagnosis and treatment decisions in discussion with the patient¹

Comparing diabetes blood tests

View the content below as a table in the [PDF version](#) (PDF, 582.72 KB) .

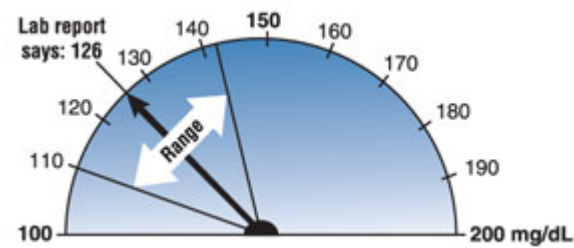
Fasting plasma glucose (FPG) test

Uses

- Screening and diagnosis of prediabetes¹ or impaired fasting glucose (IFG)
 - 100–125 mg/dL³
- Screening and diagnosis of diabetes¹
 - 126 mg/dL or higher¹
 - repeat for confirmation of diagnosis

Technical features²

- Diagnosis requires a lab test; [meter](#) results are not suitable
- Sample in morning, after 8-hour fast¹
- Sample: sodium fluoride plasma preferred
- Sample stability: low—requires processing within 30 minutes
- Sensitivity: greater than the A1C test, less than the OGTT
- Coefficient of variation: assay variability
- Biological variability



With a coefficient of variation of 5.7% (typical biological variation within the same person), an FPG test result of 126 mg/dL could indicate a true FPG of anywhere from approximately 110 to 142 mg/dL.² Image courtesy of David Aron, M.D., Louis Stokes Department of Veterans Affairs Medical Center.

Pros	Cons ²
• Low cost	• Indicates single-point blood glucose level

Pros	Cons ²
• Assay is widely available • Assay is automated	• Affected by short-term lifestyle changes, such as stress or illness • Less tightly linked to diabetes complications than A1C • Not convenient for patient or health care professional; requires fasting and scheduling a morning appointment or return visit • Diurnal variation • Sample not stable after collection • High within-patient variability • Many laboratories measure serum, which is not recommended • Inadequate standardization of assays

Oral glucose tolerance test (OGTT)

Uses

- Screening and diagnosis of prediabetes or impaired glucose tolerance (IGT)¹
 - 140 to 199 mg/dL at 2 hours¹
- Screening and diagnosis of diabetes¹
 - ≥200 mg/dL or higher at 2 hours¹
 - repeat to confirm the diagnosis

Technical features

- Performed as described by the World Health Organization (WHO), using a glucose load containing the equivalent of 75 grams of anhydrous glucose dissolved in water¹
- Sample in morning: two samples after 8-hour fast and 2 hours after glucose load⁴
- Sample stability: low—requires processing within 30 minutes
- Patients should ingest at least 150 grams/day of [carbohydrates](#) for 3 days before test²

- Sensitivity: greater than the A1C or the FPG tests
- Range of variability: 16.7%²

Pros ²	Cons ²
• Sensitive indicator of risk of developing diabetes • Early marker of impaired glucose metabolism	• Affected by short-term lifestyle changes such as stress, illness, and medications • Not convenient for patient or health care professional; requires fasting and scheduling a morning appointment or return visit • Extensive patient preparation • Sample not stable after collection • High within-patient variability • Low reproducibility • Higher cost than other tests

A1C test

The A1C test is sometimes called the hemoglobin A1C, HbA1c, glycated hemoglobin, or glycohemoglobin test.

Uses

- Screening and diagnosis of prediabetes¹
 - 5.7–6.4%¹
- Screening and diagnosis of type 2 diabetes¹
 - 6.5% or higher¹
 - repeat for confirmation of diagnosis
- Monitoring of diabetes

Technical features

- Diagnosis requires a laboratory test certified by the NGSP and standardized to the DCCT assay.¹ Some point-of-care A1C assays may be certified by the NGSP or approved by the U.S. Food and Drug Administration for diagnosis; however, they should only be considered in laboratories that are certified to perform moderate-to-high complexity tests to ensure testing proficiency.¹
- Sample at any time of day, no fasting required³
- Sample: anticoagulated whole blood
- Sample stability: superior⁵
- Sensitivity: less than the FPG test and the OGTT¹
- Coefficient of variation: for between laboratory assay variability, see [College of American Pathologists \(CAP\) survey data](#) 

Pros	Cons*
• Reflects long-term blood glucose concentration⁴ • Unaffected by acute changes in glucose levels due to stress or illness⁴ • Highly correlated with risks for complications, such as retinopathy and cardiovascular disease • Convenient for patient and health care professionals • Most stable sample after collection⁴ • Low within-patient variability⁴ • Established international standardization of lab tests¹ • Accuracy of test is monitored¹	• Lower sensitivity: identifies fewer cases of diabetes than the glucose tests¹ • Possible interference with some assay methods, resulting in falsely increased or lowered results due to some genetic hemoglobin variants (e.g., HbC, HbS, HbE, and HbD traits^{**}) and elevated fetal hemoglobin (HbF); this primarily affects people of African, Mediterranean, or Southeast Asian heritage⁶ • Altered relationship between A1C and glycemia in certain conditions¹ ◦ sickle cell disease¹ ◦ glucose-6-phosphate dehydrogenase deficiency¹ ◦ HIV¹ ◦ hemodialysis¹ ◦ recent blood loss or transfusion¹ ◦ erythropoietin therapy¹ ◦ iron-deficiency anemia¹ ◦ kidney disease⁵ ◦ liver disease⁷

Pros	Cons [*]
	• Not recommended for rapidly progressing diabetes, e.g., type 1 diabetes in children¹ • Not recommended for screening cystic fibrosis NIH—related diabetes¹ • May not be available in some laboratories/areas of the world¹ • Higher cost than glucose tests¹

* See [ngsp.org](#) [↗](#) for information on A1C interference and recommended testing methods.

** See [The A1C Test and Diabetes](#).

Random plasma glucose (RPG) test

Uses

- Diagnosis of diabetes—used only with classic symptoms of hyperglycemia or hyperglycemic crisis
 - [polyuria](#), [polydypsia](#), and unexplained weight loss
 - 200 mg/dL or higher¹

Technical features

- Sample at any time, no fasting required³
- Sample stability: low—requires processing in less than 30 min

Pros	Cons
• Convenient • Part of basic metabolic panel NIH screen	• Indicates single-point blood glucose level • Used only in symptomatic patients, not recommended for screening • Insensitive measurement • Greater within-patient variability • Affected by short-term lifestyle changes and mealtimes

Comparing diagnoses

In some people, a blood glucose test may indicate a diagnosis of diabetes even though an A1C test does not.¹

The reverse can also occur—an A1C test may indicate a diagnosis of diabetes even though a blood glucose test does not.¹

Because of these variations in test results, health care professionals should repeat tests before making a diagnosis. People with differing test results may be in an early stage of the disease, where blood glucose levels have not risen high enough to show on every test.

References

- [1] American Diabetes Association. 2. Classification and Diagnosis of Diabetes: *Standards of Medical Care in Diabetes—2020*. *Diabetes Care*. 2020;43(Suppl 1):S14–S31. doi:10.2337/dc20-S002
- [2] Sacks DB. A1C versus glucose testing: a comparison. *Diabetes Care*. 2011;34(4):518–523. doi: 10.2337/dc10-1546
- [3] Diagnosing diabetes and learning about prediabetes. American Diabetes Association. Accessed July 8, 2019. www.diabetes.org/diabetes-basics/diagnosis 
- [4] Diabetes tests and diagnosis. National Institute of Diabetes and Digestive and Kidney Diseases. Updated December 2016. Accessed March 2020. www.niddk.nih.gov/health-information/diabetes/overview/tests-diagnosis#diagnoseddiabetes
- [5] The A1C test and diabetes. National Institute of Diabetes and Digestive and Kidney Diseases. Updated April 2018. Accessed July 2019. www.niddk.nih.gov/health-information/diabetes/overview/tests-diagnosis/a1c-test
- [6] Factors that interfere with HbA1c test results. NGSP. Updated August 21, 2019. Accessed August 2020. ngsp.org/factors.asp 
- [7] Addepally NS, George N, Martinez-Macias R, Garcia-Saenz-de-Sicilia M, Kim WR, Duarte-Rojo A. Hemoglobin A1c has suboptimal performance to diagnose and monitor diabetes mellitus in patients with cirrhosis. *Digestive Diseases and Sciences* 2018;63(12):3498–3508. doi: 10.1007/s10620-018-5265-3

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